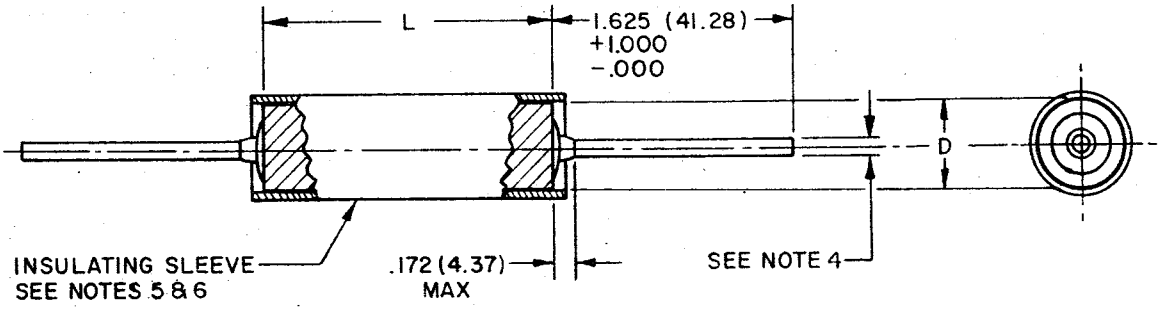
MILITARY SPECIFICATION SHEET

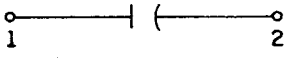
CAPACITORS, FIXED, METALLIZED (PAPER, PAPER PLASTIC, OR PLASTIC FILM) DIELECTRIC, DIRECT CURRENT, (HERMETICALLY SEALED IN METAL CASES), (INSULATED), STYLE CH09

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-18312.

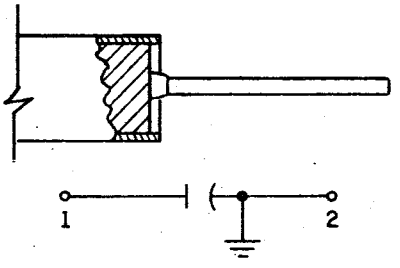
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR NEW DESIGN AFTER 8 JANUARY 1970. USE MIL-C-39022/1.





CIRCUIT DIAGRAM 1



CIRCUIT DIAGRAM 3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. See table I for additional dimensions.
4. Number 24 AWG wire for case diameters of .227 (5.77 mm) and under, number 22 AWG for case diameters .267 (6.78 mm) thru .344 (8.74 mm) and 20 AWG wire for case diameters .432 (10.97 mm) and over.
5. Plastic insulating sleeve shall be transparent: Marking shall be applied to the capacitor case.
6. Plastic insulating sleeve shall extend beyond the capacitor body, but shall not exceed .062 (.79 mm) on either end. Insulating sleeve thickness shall not exceed .016 (.41 mm).

FSC 5910

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REQUIREMENTS:

Dimensions: See figure and table I.

Case material: Nonmagnetic.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC rated voltage: See table I.

Seal:
Method 112 of MIL-STD-202, condition A.
Liquid-filled capacitors - One hour at applicable high ambient temperature ±3° C with no evidence of leakage.

Dielectric withstanding voltage:
Method 301 of MIL-STD-202.
Terminal to terminal - 200 percent of rated dc voltage.
Terminal to case (where case not a terminal) - 200 percent of rated dc voltage.

Capacitance:
Method 305 of MIL-STD-202 (for value and tolerance see table I).

Dissipation factor:
At 25° C - 1.0 percent.
At high test temperature - 2.0 percent.

Insulation resistance:
Terminal to terminal - See table III.
Terminal to case (case not a terminal) - 10,000 megohms.

Barometric pressure:
Method 105 of MIL-STD-202.
Test condition - 0.82 inches of mercury (80,000 ft).

Vibration, high frequency:
Method 204 of MIL-STD-202, condition B.

Temperature cycling: Method 102 of MIL-STD-202, condition D, except that for step 3, temperature shall be 125° C for capacitors rated at 200, 400 and 600 volts DC.

Immersion:
Method 104 of MIL-STD-202, condition C.
Insulation resistance:
Terminal to terminal - Not less than 1/3 the value specified in table III.
Terminal to case (case not a terminal) - 5,000 megohms, minimum.
Insulating sleeves - 100 megohms, minimum.
Dielectric withstanding voltage:
Insulating sleeve - 4,000 volts minimum.
Terminal to terminal - 150 percent of rated dc voltage.
Terminal to case (case not a terminal) - 200 percent of rated dc voltage.
Capacitance change - Not to exceed 10 percent of initial value.
Dissipation factor - Not to exceed 110 percent of initial requirement.

Solderability:
Method 208 of MIL-STD-202.

Terminal strength:
Method 211 of MIL-STD-202, conditions A and D.
Condition A - Applied force 4-1/2 pounds.
Condition D - 3 twists of 360 degrees.

Low temperature and capacitance change with temperature.
Conditions - 48-hour with rated voltage applied.
Capacitance change - See table III.

Life:
Qualification - Test voltage 140 percent of rated voltage, 1,000 hours at high ambient temperature.
Quality conformance - 250 hours at conditions specified for 1,000-hour test.

Dissipation factor:
At high ambient test temperature (any time 24 to 48 hours from start of test) - Not greater than 2.0 percent.
At high ambient test temperature (any time during last 48 hours of test) - Not greater than 2.5 percent.
At 25° C (after life test) - Not greater than 1.25 percent.

Capacitance:
At 25° C (after life test) - Within 10 percent of initial value.

Insulation resistance:
At 25° C (after life test) - Terminal to terminal - Not less than 1/3 value specified in table III.
Terminal to case (case not a terminal) - Not less than 5,000 megohms.

Supersession information: See table I.

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TABLE I. Dimensions, and dash numbers.

Type designation 1/	DC rated voltage	Capacitance value	Circuit diagram	Dimensions, nominal (inches) 2/		Superseded MIL-C-18312/1A and MIL-C-18312/2A dash numbers	
				L ±.062	D +.031 -.051	Capacitance tolerance ±5%	Capacitance tolerance ±10%
	Volts	µf					
CH09A1RA183-	50	0.018	1	0.750	0.175	0301	0302
CH09A3RA183-	50	0.018	3	0.688	0.175	0303	0304
CH09A1RA223-	50	0.022	1	0.750	0.175	0305	0306
CH09A3RA223-	50	0.022	3	0.688	0.175	0307	0308
CH09A1RA273-	50	0.027	1	0.750	0.175	0309	0310
CH09A3RA273-	50	0.027	3	0.688	0.175	0311	0312
CH09A1RA333-	50	0.033	1	0.750	0.175	0313	0314
CH09A3RA333-	50	0.033	3	0.688	0.175	0315	0316
CH09A1RA393-	50	0.039	1	0.750	0.195	0317	0318
CH09A3RA393-	50	0.039	3	0.688	0.195	0319	0320
CH09A1RA473-	50	0.047	1	0.750	0.195	0321	0322
CH09A3RA473-	50	0.047	3	0.688	0.195	0323	0324
CH09A1RA563-	50	0.056	1	0.750	0.235	0325	0326
CH09A3RA563-	50	0.056	3	0.688	0.235	0327	0328
CH09A1RA683-	50	0.068	1	0.750	0.235	0329	0330
CH09A3RA683-	50	0.068	3	0.688	0.235	0331	0332
CH09A1RA823-	50	0.082	1	0.750	0.235	0333	0334
CH09A3RA823-	50	0.082	3	0.688	0.235	0335	0336
CH09A1RA104-	50	0.100	1	0.750	0.235	0337	0338
CH09A3RA104-	50	0.100	3	0.688	0.235	0339	0340
CH09A1RA184-	50	0.180	1	0.874	0.312	0341	0342
CH09A3RA184-	50	0.180	3	0.812	0.312	0343	0344
CH09A1RA224-	50	0.220	1	0.874	0.312	0345	0346
CH09A3RA224-	50	0.220	3	0.812	0.312	0347	0348
CH09A1RA274-	50	0.270	1	0.874	0.312	0349	0350

See footnote at end of table I.

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TABLE I. Dimensions, and dash numbers. - Continued

Type designation 1/	DC rated voltage	Capacitance value	Circuit diagram	Dimensions, nominal (inches) 2/		Superseded MIL-C-18312/1A and MIL-C-18312/2A dash numbers	
				L ±.062	D +.031 -.051	Capacitance tolerance	
						±5%	±10%
	Volts	µf					
CH09A3RA274-	50	0.270	3	0.812	0.312	0351	0352
CH09A1RA334-	50	0.330	1	0.874	0.312	0353	0354
CH09A3RA334-	50	0.330	3	0.812	0.312	0355	0356
CH09A1RA394-	50	0.390	1	1.125	0.312	0357	0358
CH09A3RA394-	50	0.390	3	1.062	0.312	0359	0360
CH09A1RA474-	50	0.470	1	1.125	0.312	0361	0362
CH09A3RA474-	50	0.470	3	1.062	0.312	0363	0364
CH09A1RA564-	50	0.560	1	1.125	0.400	0365	0366
CH09A3RA564-	50	0.560	3	1.062	0.400	0367	0368
CH09A1RA684-	50	0.680	1	1.125	0.400	0369	0370
CH09A3RA684-	50	0.680	3	1.062	0.400	0371	0372
CH09A1RA824-	50	0.820	1	1.125	0.400	0373	0374
CH09A3RA824-	50	0.820	3	1.062	0.400	0375	0376
CH09A1RA105-	50	1.0	1	1.125	0.400	0377	0378
CH09A3RA105-	50	1.0	3	1.062	0.400	0379	0380
CH09A1RA155-	50	1.5	1	1.125	0.500	0381	0382
CH09A3RA155-	50	1.5	3	1.062	0.500	0383	0384
CH09A1RA225-	50	2.2	1	1.125	0.562	0385	0386
CH09A3RA225-	50	2.2	3	1.062	0.562	0387	0388
CH09A1RA275-	50	2.7	1	1.375	0.562	0389	0390
CH09A3RA275-	50	2.7	3	1.312	0.562	0391	0392
CH09A1RA335-	50	3.3	1	1.375	0.562	0393	0394
CH09A3RA335-	50	3.3	3	1.312	0.562	0395	0396
CH09A1RA395-	50	3.9	1	1.625	0.562	0397	0398
CH09A3RA395-	50	3.9	3	1.562	0.562	0399	0400

See footnote at end of table I.

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TABLE I. Dimensions, and dash numbers. - Continued

Type designation 1/	DC rated voltage	Capacitance value	Circuit diagram	Dimensions, nominal (inches) 2/		Superseded MIL-C-18312/1A and MIL-C-18312/2A dash numbers	
				L ±.062	D +.031 -.051	Capacitance tolerance	
						±5%	±10%
	Volts	µf					
CH09A1RA475-	50	4.7	1	1.875	0.562	0401	0402
CH09A3RA475-	50	4.7	3	1.812	0.562	0403	0404
CH09A1RA565-	50	5.6	1	1.625	0.670	0405	0406
CH09A3RA565-	50	5.6	3	1.562	0.670	0407	0408
CH09A1RA685-	50	6.8	1	1.875	0.670	0409	0410
CH09A3RA685-	50	6.8	3	1.812	0.670	0411	0412
CH09A1RA106-	50	10.0	1	1.875	0.750	0413	0414
CH09A3RA106-	50	10.0	3	1.812	0.750	0415	0416
CH09A1RA126-	50	12.0	1	2.375	0.750	0417	0418
CH09A3RA126-	50	12.0	3	2.312	0.750	0419	0420
						±10%	±20%
CH09A1NC104-	200	0.10	1	0.844	0.312	0421	0422
CH09A3NC104-	200	0.10	3	0.781	0.312	0423	0424
CH09A1NC224-	200	0.22	1	1.125	0.312	0425	0426
CH09A3NC224-	200	0.22	3	1.062	0.312	0427	0428
CH09A1NC474-	200	0.47	1	1.125	0.400	0429	0430
CH09A3NC474-	200	0.47	3	1.062	0.400	0431	0432
CH09A1NC105-	200	1.0	1	1.125	0.562	0433	0434
CH09A3NC105-	200	1.0	3	1.062	0.562	0435	0436
CH09A1NC155-	200	1.5	1	1.844	0.562	0437	0438
CH09A3NC155-	200	1.5	3	1.781	0.562	0439	0440
CH09A1NC225-	200	2.2	1	1.844	0.562	0441	0442
CH09A3NC225-	200	2.2	3	1.781	0.562	0443	0444
CH09A1NC335-	200	3.3	1	1.875	0.670	0445	0446
CH09A3NC335-	200	3.3	3	1.812	0.670	0447	0448
CH09A1NC475-	200	4.7	1	1.875	1.000	0449	0450

See footnote at end of table I.

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TABLE I. Dimensions, and dash numbers. - Continued

Type designation 1/	DC rated voltage	Capacitance value	Circuit diagram	Dimensions, nominal (inches) 2/		Superseded MIL-C-18312/1A and MIL-C-18312/2A dash numbers	
				L ±.062	D +.031 -.051	Capacitance tolerance ±10%	±20%
	Volts	µf					
CH09A3NC475-	200	4.7	3	1.812	1.000	0451	0452
CH09A1NC685-	200	6.8	1	1.875	1.000	0453	0454
CH09A3NC685-	200	6.8	3	1.812	1.000	0455	0456
CH09A1NC106-	200	10.0	1	2.375	1.000	0457	0458
CH09A3NC106-	200	10.0	3	2.312	1.000	0459	0460
CH09A1NC126-	200	12.0	1	2.625	1.000	0461	0462
CH09A3NC126-	200	12.0	3	2.562	1.000	0463	0464
CH09A1NE473-	400	0.047	1	1.125	0.312	0465	0466
CH09A3NE473-	400	0.047	3	1.062	0.312	0467	0468
CH09A1NE104-	400	0.10	1	1.125	0.400	0469	0470
CH09A3NE104-	400	0.10	3	1.062	0.400	0471	0472
CH09A1NE224-	400	0.22	1	1.125	0.562	0473	0474
CH09A3NE224-	400	0.22	3	1.062	0.562	0475	0476
CH09A1NE474-	400	0.47	1	1.625	0.562	0477	0478
CH09A3NE474-	400	0.47	3	1.562	0.562	0479	0480
CH09A1NE105-	400	1.0	1	1.875	0.750	0481	0482
CH09A3NE105-	400	1.0	3	1.812	0.750	0483	0484
CH09A1NE225-	400	2.2	1	1.875	1.000	0485	0486
CH09A3NE225-	400	2.2	3	1.812	1.000	0487	0488
CH09A1NE335-	400	3.3	1	2.625	1.000	0489	0490
CH09A3NE335-	400	3.3	3	2.562	1.000	0491	0492
CH09A1NF103-	600	0.01	1	0.812	0.312	0493	0494
CH09A3NF103-	600	0.01	3	0.750	0.312	0495	0496
CH09A1NF223-	600	0.022	1	0.812	0.312	0497	0498
CH09A3NF223-	600	0.022	3	0.750	0.312	0499	0500

See footnote at end of table I.

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TABLE I. Dimensions, and dash numbers. - Continued

Type designation 1/	DC rated voltage	Capacitance value	Circuit diagram	Dimensions, nominal (inches) 2/		Superseded MIL-C-18312/1A and MIL-C-18312/2A dash numbers	
				L ±.062	D +.031 -.051	Capacitance tolerance ±10%	±20%
	Volts	μf					
CH09A1NF473-	600	0.047	1	1.125	0.400	0501	0502
CH09A3NF473-	600	0.047	3	1.062	0.400	0503	0504
CH09A1NF104-	600	0.10	1	1.125	0.500	0505	0506
CH09A3NF104-	600	0.10	3	1.062	0.500	0507	0508
CH09A1NF224-	600	0.22	1	1.375	0.562	0509	0510
CH09A3NF224-	600	0.22	3	1.312	0.562	0511	0512
CH09A1NF474-	600	0.47	1	1.625	0.670	0513	0514
CH09A3NF474-	600	0.47	3	1.562	0.670	0515	0516
CH09A1NF105-	600	1.0	1	1.844	1.000	0517	0518
CH09A3NF105-	600	1.0	3	1.781	1.000	0519	0520
CH09A1NF155-	600	1.5	1	1.875	1.000	0521	0522
CH09A3NF155-	600	1.5	3	1.812	1.000	0523	0524
CH09A1NF225-	600	2.2	1	2.625	1.000	0525	0526
CH09A3NF225-	600	2.2	3	2.562	1.000	0527	0528

1/ Complete type designation will include an additional symbol for capacitance tolerance.
2/ Dimensions shown are for bare case. (see figure)

TABLE II. Millimeter equivalents of decimal inches.

Inches	MM	Inches	MM
.031	.79	.874	22.20
.051	1.30	1.000	25.40
.062	1.57	1.062	26.97
.175	4.45	1.125	28.58
.195	4.95	1.312	33.32
.235	5.97	1.375	34.93
.312	7.92	1.562	39.67
.400	10.16	1.625	41.28
.500	12.70	1.781	45.24
.562	14.27	1.812	46.02
.670	17.02	1.844	46.84
.688	17.48	1.875	47.63
.750	19.05	2.312	58.72
.781	19.84	2.375	60.33
.812	20.62	2.562	65.07
.844	21.44	2.625	66.68

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TABLE III. Characteristics.

Operating temperature range	-55 to +125 ±3° C		-55 to +85 ±3° C
DC voltage rating (volts)	200	400 600	50
High ambient test temperature	125° ±3° C	125° ±3° C	85° ±3° C
Low ambient test temperature	-55° ±3° C	-55° ±3° C	-55° ±3° C
Megohms x microfarads:			
At 25° C	2,000 min	2,000 min	25,000 min
At high ambient test temperature	10 min	40 min	500 min
Insulation resistance (megohms):			
At 25° C 1/	12,000	12,000	50,000
At high ambient test temperature 1/	150	600	5,000
Capacitance change at low temperature	-15% max	-15% max	-8% max
Life test voltage (% rated voltage)	140%	140%	140%

1/ Need not exceed.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - EC
Air Force - 11, 17, 85

User activities:
Army - None
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

(Project 5910-0968-2)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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